HARTINGTON IP

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FREEDOM-TO-OPERATE SEARCH REPORT

Subject: RapidCure out-of-autoclave (OOA) CFRP cure process

Client: Halstead Aerospace Limited

Field reviewed: eVTOL primary airframe structures (MTOM ≤ 3,175 kg)

Report ref: HIP-FTO-2024-114

Date: 21 October 2024

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1. Scope

This report assesses the freedom to operate of the RapidCure process within the eVTOL field of use. Searches were conducted across EPO, USPTO, WIPO and GB registers for in-force patents disclosing rapid out-of-autoclave curing of carbon-fibre reinforced polymer structures, heated-tooling thermal control, and vacuum-bag consolidation.

2. Halstead patent family reviewed

Publication	Jurisdiction	Status	Expiry
GB2578941B	UK	Granted, in force	12 Mar 2039
EP3712198B1	EP (UK,FR,DE,IT)	Granted, in force	12 Mar 2039
US11,247,883B2	US	Granted, in force	04 Aug 2039
PCT/GB2023/051877	PCT	Pending (national phase)	–

3. Third-party patents of potential relevance

Two third-party patents were identified as warranting analysis:

Patent	Proprietor	Relevance	Assessment
EP3105112B1	Mitsuwa Composites K.K.	Vacuum-bag consolidation of prepreg laminates	Does not read on: RapidCure uses two-stage dwell profile ou
US10,894,367B2	Toulon Aerospace SAS	Closed-loop heated-tooling temperature control	Does not read on: claims limited to single-zone control; Rapid

4. Conclusion

On the basis of the searches conducted, we consider the freedom-to-operate risk for the RapidCure process within the eVTOL field of use to be LOW. We identified no in-force third-party patent whose claims, properly construed, read onto the RapidCure process as practised. This opinion is limited to the patents identified by the searches and to the field of use stated above.

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